

The Orthogonal Decomposition

splitting a vector into a parallel part and an orthogonal part

Lecture 2 notes

1 Statement (Axler 6.13)

Let $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{v} \neq \mathbf{0}$. Then $\mathbf{u}$ splits *uniquely* into a part along $\mathbf{v}$ and a part orthogonal to $\mathbf{v}$:

$$\mathbf{u} = \underbrace{\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v}}_{\text{parallel part} = \text{proj}_{\mathbf{v}} \mathbf{u}} + \underbrace{\left(\mathbf{u} - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} \right)}_{\text{orthogonal part} = \mathbf{w}}.$$

Equivalently, $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ with $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ and

$$c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}.$$

2 Why the scalar is what it is

The scalar is *not guessed* — it is *solved for* by the single requirement that the leftover part be orthogonal to $\mathbf{v}$. We want

$$\langle \mathbf{u} - c\mathbf{v}, \mathbf{v} \rangle = 0.$$

Using linearity in the first slot and $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$:

$$\langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = 0 \implies c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}.$$

Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\|^2 \neq 0$ so the division is legal, and the value of c — hence the whole decomposition — is *unique*.

3 The picture

4 Pythagoras applies

Because the two pieces are orthogonal, their norms combine by Pythagoras:

$$\|\mathbf{u}\|^2 = \|\text{proj}_{\mathbf{v}} \mathbf{u}\|^2 + \|\mathbf{w}\|^2.$$

This is the engine our triangle-inequality proof used, and it is exactly how Axler proves the Cauchy–Schwarz inequality in one line: starting from this decomposition for fixed $\mathbf{v}$, Pythagoras gives

$$\|\mathbf{u}\|^2 \geq \left\| \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} \right\|^2 = \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^4} \|\mathbf{v}\|^2 = \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2},$$

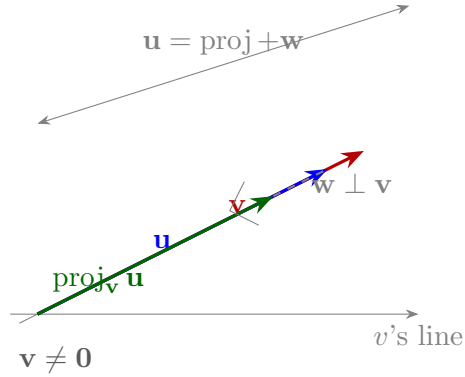


Figure 1: The orthogonal decomposition: $\mathbf{u}$ is the sum of its projection onto $\mathbf{v}$'s direction (solid) and its orthogonal part $\mathbf{w}$ (dashed, drawn with a right-angle mark to show $\langle \mathbf{w}, \mathbf{v} \rangle = 0$).

and multiplying by $\|\mathbf{v}\|^2$ and taking square roots yields

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

the Cauchy–Schwarz inequality — equality iff the orthogonal part $\mathbf{w} = \mathbf{0}$, i.e. iff $\mathbf{u}$ is a scalar multiple of $\mathbf{v}$.

5 Worked example

Take $\mathbf{u} = (2, 1)$ and $\mathbf{v} = (1, \sqrt{3})$ in $\mathbb{R}^2$ with the Euclidean inner product. Then

$$\langle \mathbf{u}, \mathbf{v} \rangle = 2 \cdot 1 + 1 \cdot \sqrt{3} \approx 3.732, \quad \|\mathbf{v}\|^2 = 1^2 + (\sqrt{3})^2 = 4,$$

so $c = 3.732/4 \approx 0.933$. Hence

$$\begin{aligned} \text{proj}_{\mathbf{v}} \mathbf{u} &= 0.933 (1, \sqrt{3}) \approx (0.933, 1.616), \\ \mathbf{w} &= \mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u} \approx (1.067, -0.616). \end{aligned}$$

Check orthogonality:

$$\langle \text{proj}_{\mathbf{v}} \mathbf{u}, \mathbf{w} \rangle \approx 0.933 \cdot 1.067 + 1.616 \cdot (-0.616) \approx 0.996 - 0.995 \approx 0.$$

Check Pythagoras:

$$\|\text{proj}_{\mathbf{v}} \mathbf{u}\|^2 + \|\mathbf{w}\|^2 \approx 3.48 + 1.52 \approx 5.00 = 2^2 + 1^2 = \|\mathbf{u}\|^2.$$

6 Where this leads

The orthogonal decomposition is the workhorse behind everything that follows: it proves Cauchy–Schwarz (as above), it constructs orthonormal bases via the *Gram–Schmidt* process (subtract each new vector's projections onto the already-found basis vectors), and it solves least-squares minimization problems. It is the same idea we drew for the triangle inequality — now a precise, reusable tool.